Portable Microplastic Survey Boat

Critical Components, Payload, Power and Two-Hour Mission Feasibility

Year 3 Engineering Group Project — preliminary systems design

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Abstract

This independent engineering report translates the requirement to produce a spatial microplastic concentration heat map across a 20 m × 20 m water area within about two hours into a first-pass hull, propulsion, autonomy, pump, communications, and energy architecture. Manufacturer-rated values are cited separately from illustrative engineering assumptions. It does *not* establish an actual motor payload rating, boat top speed, sensor selectivity or two-hour field performance. The companion methodology review contains the scientific measurement screening.

1 Mission requirements, operating modes and acceptance criteria

A 400 m^2 water survey must yield georeferenced observations and a rendered 2D concentration or explicitly provisional particle proxy map inside 120 minutes. Track A measures aboard a moving or stopped boat; Track B collects known-volume labelled water at 16 illustrative grid locations for portable shoreline analysis. Remote operator retains positive control and a safe manual retrieval plan. The shoreline/boat route, local current, obstacle map, GNSS performance, wind and water depth are not specified and require field reconnaissance.

Table 1: Proposed acceptance gates; numerical criteria are design targets, not published hardware performance.

Dimension	Acceptance demonstration	Consequence if failed
Geospatial output	20×20 m map, original locations, timestamps, size limits, data-quality legend	Map fails stakeholder purpose
Timeliness	First deployment to completed map within ≤ 120 min	Reduce stations or improve throughput
Measurement credibility	Controlled known-particle trials, river interferents, measured sample volume, blanks	Label "suspected MP-like particles" or calibrated proxy
Navigation	Confirm track spacing and positional error; manual RC override	No autonomous close-range operation
Flotation	Float full wet load with reserve buoyancy and stable freeboard	Increase hull displaced volume
Energy	120 min at *measured* average load, BMS/fuse adequate for transients	More battery, lower draw or mission redesign

2 Architecture and critical component list

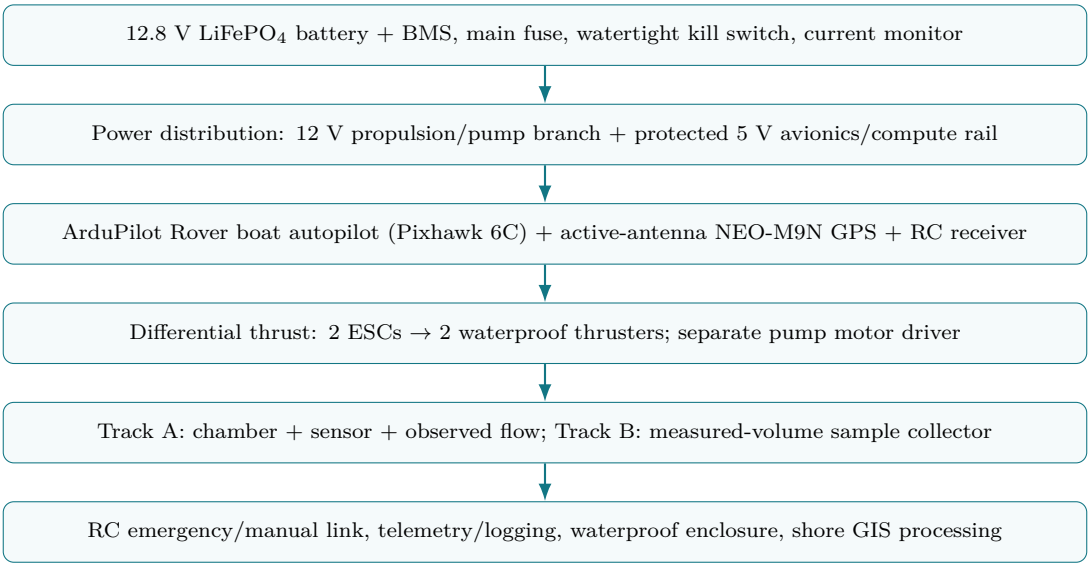


Figure 1: Block diagram (logical, not complete wiring schematic). RC safety and boat-side logs must not depend on the data-telemetry link.

Table 2: Prototype system purchase priorities; masses are budget allowances unless specified by manufacturers.

Critical system	Candidate hardware	Procurement/check before commitment
Hull/buoyancy	2 sealed cylindrical pontoons, 0.90 m long ×0.20 m OD, rigid platform	Measure true sealed displacement, frame strength and wet load
Propulsion	2 Blue Robotics T200 + bidirectional ESCs	12 V compatible, guards, differential steering and sufficient BMS peak current [1]
Autopilot	Holybro Pixhawk 6C, ArduPilot Rover firmware	Supported hardware; boat mode uses `FRAME_CLASS=2` [2], [3]
RC link	RadioMaster Pocket 2.4 GHz ELRS with matched serial receiver	Manually test kill/hold/failsafe; ensure regional radio compliance [4]
Telemetry	2.4 GHz Wi-Fi for near-bank test; optional compatible SiK pair	MAVLink/logs and onboard SD backup, independent of RC [5]
GNSS	u-blox **NEO-M9N-00B** module + compatible powered active L1 antenna/compass unit	Verify receiver board has antenna bias circuitry, magnetometer and correct autopilot cable [6], [7]
Sampling	SEAFLO 21-Series 12 V diaphragm pump + calibrated flowmeter	1.2 GPM nameplate is open flow; flow through filters may be lower [8]
Power	12.8 V 20 Ah $LiFePO_4$, integrated BMS, $LiFePO_4$ charger	Peak current and fuse coordination; avoid powering motor from flight-controller rail [9], [10]
Safety	Sealed electronics, buoyant tether/recovery aid, prop guards, water-leak detector	Safety switch, loss-of-RC configuration, emergency hand retrieval

3 Radio transmission: RC control versus basic telemetry

Table 3: Two distinct links; ranges depend on water-level antenna placement and site conditions.

Function	Example	Design consequence
Manual RC override	RadioMaster Pocket, ExpressLRS 2.4 GHz receiver; CRSF serial input	Priority for emergency stop/hold/manual steering; independently test receiver loss, binding and channel assignment [4]
Basic data transmission	Short-range 2.4 GHz Wi-Fi to shore, or paired SiK MAVLink radio	Send time/GPS/battery/QC/concentration updates; do not stream high-resolution raw video unless capacity verified [5]
Onboard resilience	SD logging in autopilot or companion computer	Store raw sensor files and logs even when telemetry is lost
Frequency compliance	Malaysia MCMC Class Assignment No. 2 of 2025	Confirm permitted band, module approvals and transmit power before using 433/868/915 MHz [11]

Proposed telemetry packet (not a standard specification): mission ID, timestamp, latitude, longitude, fix quality, battery V/A, motor duty, flow rate, sample ID, counted events, confidence and fault bits. At one compact packet per second, average data rate should be small; verify actual payload and overhead. Record all high-bandwidth images locally.

4 Autopilot versus in-house control board

Table 4: Drone flight controller reuse is supported, but a generic drone ECU is not automatically boat-ready.

Criterion	Pixhawk 6C / ArduPilot Rover	In-house ESP32 / STM32 MCU
Boat software	Official boat variant uses `FRAME_CLASS=2`; Rover supports GPS waypoints, Auto, Guided, RTL and Loiter [3]	Must implement navigation, steering loops, state estimation and failsafes
Hardware	STM32H743, multiple PWM, serial/CAN/GPS interfaces; supported ArduPilot releases [2], [12]	Low-cost and flexible I/O, but architecture is student-developed
Twin-motor drive	Servo-output differential thrust configuration supported with correct vehicle setup	Two ESC PWM signals; differential thrust and mixing coded manually
Mission risk	Mature navigation stack; test firmware, prop guards and mode switching	More coding and test burden, higher risk under two-hour deadline
Recommended division	Safety navigation, waypoint execution and RC override	Sampling scheduler, sensor data, flowmeter and CSV telemetry

The supplied self-driving boat tutorial is a useful build example, but *the governing technical confirmation is ArduPilot's official boat configuration documentation.* Parameter/firmware compatibility must be checked on the chosen board. Set motor outputs, steering and failsafes on the actual boat, not by copying an aircraft mixer.

5 GPS: active antenna, receiver model and mapping implications

Specify the **NEO-M9N-00B** multiconstellation receiver (or Holybro M9N GPS/compass kit with verified module revision). A breakout with *active-antenna bias* is required if using a powered L1 GNSS patch; a bare antenna connector does not guarantee power to an active antenna. u-blox documents support for active antennas, and SparkFun offers NEO-M9N breakouts with external antenna connectors [6], [7]. Representative manufacturer positioning accuracy is roughly 1.5 m horizontal CEP under specified conditions [6]. This is substantial relative to proposed 5 m grid spacing; repeat ground-point checks, keep GNSS well above the waterline, separate from motors, and log actual fix quality. If sub-metre mapping is essential, upgrade to an RTK-capable *different* receiver and surveyed reference corrections rather than claiming the NEO-M9N provides RTK.

6 Motor comparison: thrust is not hull payload capacity

An electric thruster's static thrust (kgf) balances *horizontal drag*, not the vertical weight that the hull must float. The allowable carried mass is governed primarily by sealed displacement and stability. Therefore the table compares **manufacturer thrust and full-throttle electrical load**; it cannot supply a legitimate "maximum payload kg" without hull geometry, drag and target speed.

Table 5: Manufacturer-rated propulsion comparison. Full-throttle figures must not be interpreted as average cruise consumption.

Motor option	Full forward thrust	Full-load electrical at stated V	Proposed load class*	Implication
2 × T200 (12 V) [1]	3.71 kgf *each*	17 A, 205 W *each*	Prototype 15–25 kg	Good steering authority, *requires throttle limit and peak-current validation*
2 × T500 (12 V) [13]	6.0 kgf *each*	16.9 A, 202 W *each*	Heavier hulls, >25 kg	Larger/heavier, likely overkill for sheltered 20-m survey
Minn Kota Endura C2 30 (12 V) [14]	30 lbf ≈ 13.6 kgf total	up to 30 A, ≈ 360 W	Larger slow platform	Manual tiller/steering adaptation and weight; no differential drive by default

*The "load class" is a proposed *prototype trial mass*, not a rated motor payload or a verified towing capacity. Select actual motors after tow/drag, current and turning tests.

Measured-load test: add 5-kg ballast steps to the completed hull, measure straight-line speed, DC power, steering response, stopping, and recovery. Record operating powers at e.g. 0.15, 0.25 and 0.40 m/s rather than fabricating a vendor speed–payload curve.

7 Hydrostatics and expected payload mass

7.1 Twin-pontoon displacement sizing

For two sealed circular cylinders, $L = 0.90$ m and $D = 0.20$ m:

$$V_{\max} = 2\pi(D/2)^2 L = 0.05655\,\mathrm{m}^3 \approx 56.55\,\mathrm{L}.$$

In fresh water, theoretical maximum fully submerged displaced water mass is approximately 56.55 kg. Using no more than **60% of that displacement as an initial working buoyancy budget** gives about **33.93 kg** supporting total vessel and payload. This is not a hull stability certification: caps, leaks, trim, loading location and waves may reduce available margin.

Table 6: Illustrative all-up wet mass budget (`data/payload_estimate.csv`); validate every line after procurement.

Item	kg	Item	kg
Two 0.9-m pontoons	2.30	Frame and enclosure	2.30
2 T200 + 2 ESCs	1.15	Autopilot/GPS/radios	0.45
Pump, tubing, filters	1.27	Camera and compute	0.40
12.8 V 20 Ah $LiFePO_4$	2.40	Wiring/fuses/extra sensors	0.90
Empty bottles	0.80	Eight litres retained water	8.00
Build contingency	2.00		
Total wet mass	**22.0**	Working buoyancy budget	**33.93**

Important: Track B with *16 litres retained* instead of 8 adds another 8 kg, raising the vessel to about **30.0 kg**. It is still under the 33.93-kg illustrative limit but with poor margin for wave, trim and modelling error. An approach with small bottles, fewer stations or progressive sample unloading is preferable to carrying 16 L simultaneously.

8 Water-pump selection and operating load

Table 7: Published pump specifications. Quoted open-flow rates are not actual filtered flow; record true Q under backpressure, intake lift and fouling.

Model	Supply	Flow rating	Electrical load	Use/risk
Adafruit 1150 peristaltic [15]	12 V	up to 0.10 L/min	0.2–0.3 A = 2.4–3.6 W	Low contamination contact; too slow for many 1-L stations
SEAFLO 21 SFDP1-012-035-21 [8]	12 V	1.2 GPM ≈ 4.5 L/min	4.0 A max ≈ 48 W	Candidate collection pump; intermittent duty; measure real flow at filter
SEAFLO 500 GPH bilge SFBP1-G500-01 [16]	12 V	500 GPH nominal	1.6 A nominal; 4 A max	High throughput, low head, not precision metering; needs flowmeter
SEAFLO 33-Series SFWP1-030-070-33 [17]	12 V	3.0 GPM ≈ 11.3 L/min	18 A max ≈ 216 W	Heavy duty, high current; unnecessary for initial small-scale sampling

Chosen design basis: SEAFLO 21 for Track B filling or a bypass-controlled Track A stream, with *measured* flow ≥ 1.5 L/min for Track B in the timing example. Sensitive microchannels may need a *separate low-flow metering stage*; do not force unfiltered river water through them. An inline flowmeter plus calibration against graduated water volume is compulsory. Provide coarse guard, cleanable strainers and a properly routed overflow/backflush. Some pump-contact materials are polymers: use field/process blanks and assess sample contamination.

9 Power architecture, management, charging and endurance

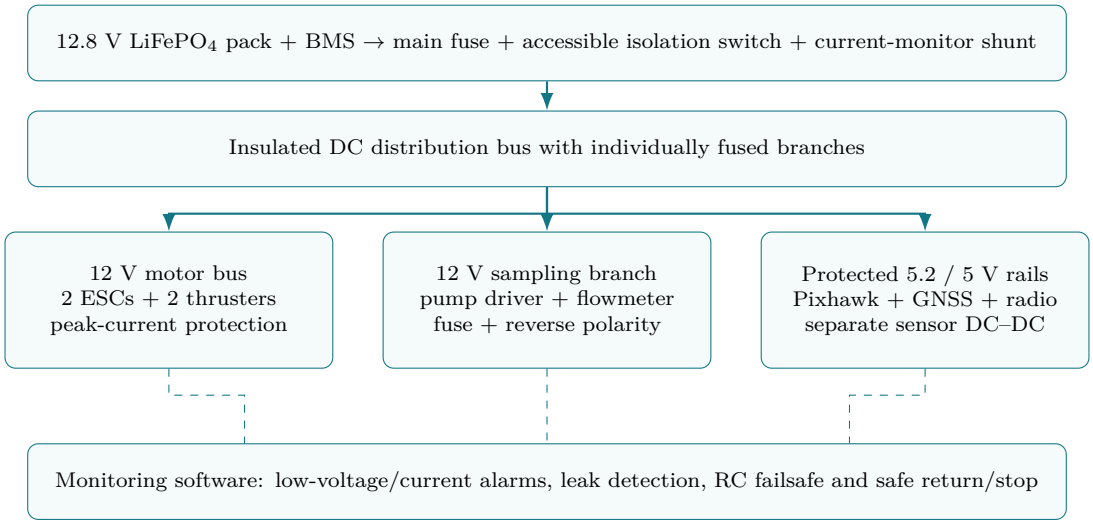


Figure 2: Power-distribution concept with genuinely parallel fused motor, pump and logic branches. Dashed paths denote monitoring/fault signalling rather than physical series wiring.

Controller power: The Holybro PM02D offers regulated 5.2 V up to 3 A from a 2S–12S input, but is *not* a high-current supply for servos, pump or motors [10]. Verify analog/digital module type against the selected Pixhawk connector. Use fused motor and pump circuits; fuses protect wires, not simply the device. **Battery charging:** A representative Bioenno 12 V 20 Ah $LiFePO_4$ pack includes a protection module with 30 A continuous discharge and specifies a suitable 14.6 V charger (manufacturer example 4 A) [9]. Do not charge $LiFePO_4$ with an unmatched lead-acid or LiPo charger. Charge in a dry supervised location, disconnected from the boat loads, with supplier-approved charger and appropriate temperature constraints.

Table 8: Averaged survey-power allowances at the DC bus: *engineering assumptions*, not manufacturer measurements.

Component / combined function	Track A W	Track B W
Two thrusters at low-speed survey (combined)	50	30
Autopilot, GNSS, RC receiver	6	6
Telemetry / logging	4	2
Camera/computer	9	0
Pump, *duty-averaged*	12	6
Conversion/misc. margin	7	4
Estimated average	**88**	**48**

$$E_{\text{nom}} = 12.8 \times 20 = 256\,\text{Wh}, \qquad E_{\text{usable}} = 0.80 \times 256 = 204.8\,\text{Wh}, \qquad t = E_{\text{usable}}/P_{\text{average}}.$$

Table 9: Illustrative 12.8 V, 20 Ah battery operating time: energy-only estimates, ignoring rate/thermal limits and unmodelled loads. Source: `data/power_endurance.csv`.

Scenario	Avg W	h	Energy for 2h	2h energy gate
Track B economical	48	4.27	96 Wh	Pass
Track A planned	88	2.33	176 Wh	Pass (narrow)
Track A difficult water	125	1.64	250 Wh	Fail
High-drive manoeuvres	180	1.14	360 Wh	Fail
Full-throttle dual T200 at 12 V + loads	430	0.48	860 Wh	Fail; BMS also inadequate

Critical current warning: Two T200 thrusters at maximum 12 V load could draw approximately $2 \times 17 = 34$ A before other components. The representative Bioenno 20 Ah pack is rated 30 A continuous; full power on both

thrusters is therefore **not permissible** with this battery/BMS. Throttle-limit the ESCs or select a pack with a verified higher discharge rating, suitably rated wires and protection. The 88 W Track A estimate requires validation with an inline wattmeter on the water, including startup and snagged-propeller cases. [1], [9]

10 Time budget and survey choices

Track A illustrative path: five 20-m lanes plus four 5-m lateral moves gives 120 m. At an assumed 0.25 m/s the ideal transit is 8 min. A 60-min effective sensing period plus 25 min setup, turns, recovery, QC and map generation totals 93 min. A 0.50-L/min fully interrogated flow yields 30 L in the sensing period, *not* 30 L per map cell. Sensor sampling time and intake-delay calibration matter.

Track B illustrative workflow: 16 stations, 1 L per site, effective pumping 1.5 L/min, 1.2 min/station for handling and 3 min/station portable image processing; add 18 min survey movement/setup/mapping. The total is about 95.9 min. Changing image time to 5 min/site raises it above 120 min. Field concentration and sampling statistics impose a stricter constraint: at around 3.12 particles/L, a 1-L sample expects only 3.12 particles, equivalent to 57% approximate Poisson standard error *before* sensor losses. The first demonstration should specify a minimum reporting volume and show uncertain/insufficient-count cells rather than inventing a smooth precise heat map.

11 Risk controls, procurement order and decision

1. **Float test first:** assemble sealed pontoons, weigh all components, add ballast to 22 and 30 kg, check reserve freeboard, trim and wave splash under stationary and moving load.
2. **Then drive/power test:** two T200 and ESCs (or lower-cost alternative only with measured equivalent performance), current limit and speed tests at full mass. Inspect for overheating and waterproof cable glands.
3. **Autopilot and RC:** flash appropriate Rover firmware, configure boat class and motor layout, test manual/RTL/Loiter and safe response to link or GNSS loss while tethered.
4. **Sample circuit:** bench-test the selected pump's *actual* flow through chosen strainers/filter and tubing at target depth; verify sampled volume and cleaning blanks.
5. **Timing and credibility:** time full 16-sample shoreline chain or Track A effective observed-volume throughput; reject any concept unable to create an auditable map within 120 min.
6. **Procurement and local radio:** request quotations from university-registered electronics/hardware suppliers, confirm Malaysia radio approval, battery transport/charging and replacement parts.

Provisional bill-of-materials decision: two 0.9-m × 0.20-m sealed pontoons, two T200 thrusters with compatible reversible ESCs, Pixhawk 6C running ArduPilot Rover boat firmware, external-active-antenna NEO-M9N (GPS/compass integration), ELRS 2.4-GHz RC control, separate low-bandwidth telemetry and local logs, SEAFLO 21 pump plus flowmeter, and 12.8 V 20 Ah $LiFePO_4$ with correct charger and managed current limits. *No full 16-L sample collection without revising the wet-mass, freeboard and storage plan.* The expected 22-kg wet all-up budget is an engineering starting value, not an independently confirmed built boat.

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